# RuleCat

## Secure COBOL Rule Extraction

Turn legacy COBOL into a searchable, traceable business-rule catalog

Team Aparajitas | IBM Z Datathon 2026

# The Problem

- Banks run on decades-old COBOL that nobody can fully explain.
- Senior mainframe experts are retiring.
- Proprietary code cannot be pasted into public AI tools.
- Result: modernization and audits are slow, costly and risky.

# Our Solution

- An on-premises tool that parses COBOL.
- A local LLM explains each rule in plain English.
- Every rule links to its exact source lines.
- Doubtful rules are flagged for human review.
- Nothing leaves the bank's infrastructure.

# How It Works




- Parser finds IF, EVALUATE and COMPUTE logic, grouped by paragraph.
- Flags catch missing ELSE, missing WHEN OTHER, GO TO, and arithmetic without ROUNDED.
- Output is a searchable rules.json with line references.

# Demo

- Input: acctcalc.cbl, a sample banking program.
- Output: 4 rules (interest tiers, interest calculation, fee, loan eligibility), each with line numbers and flags.
- Search: python rulecat.py search interest fee

```
C:\Users\VANSHIKA\cobol-rulecat>python rulecat.py extract sample/acctcalc.cbl
4 rules saved to rules.json

[R001] CALC-INTEREST lines 20-27
  This business rule calculates a rate based on the balance, with higher balances resulting in higher rates (0.045 for balances over 100,000, 0.030 for bala
nces over 10,000, and a lower rate for balances under 10,000).

[R002] CALC-INTEREST lines 28-28  ⚠arithmetic without ROUNDED
  This code calculates the interest earned by dividing the interest rate by 12 and then multiplying the balance by the resulting rate, storing the result in
 the variable WS-INTEREST.

[R003] CALC-FEE lines 30-34
  The code checks if the current balance is less than 1000 and, if so, sets the fee to 12.50, otherwise sets it to 0.

[R004] CHECK-LOAN lines 36-38  ⚠no ELSE branch
  This rule checks if a person is at least 18 years old and has an income above $25,000, then sets a flag 'ELIGIBLE' to 'Y' if the conditions are met.

C:\Users\VANSHIKA\cobol-rulecat>
```

# Who Benefits

- Modernization leads: discovery is the costliest, riskiest phase.
- Compliance and audit teams: rules traced to exact source lines.
- Business analysts: change a fee without reading COBOL.
- Mainframe operations: less dependence on a few key experts.
- Integrators and IBM partners: faster client engagements.

# IBM Z / LinuxONE Fit

- Deployment target: a LinuxONE Linux guest, next to the mainframe estate.
- Keeps source on-premises, using hardware encryption and workload isolation.
- Plan: s390x-compatible runtimes (e.g. llama.cpp), and evaluate on-chip AI acceleration.

*Prototype is portable Python + Ollama; s390x testing is planned.*

# Data and Validation

- Public COBOL samples (GnuCOBOL, Open Mainframe Project).
- Synthetic banking programs with known correct rules.
- Hand-labeled ground truth of 30-50 rules to measure accuracy.
- Synthetic transactions to check rules match actual behavior.

*Benchmark and ground-truth set are planned next steps.*

# Roadmap and Impact

| NOW | NEXT | LATER |
| --- | --- | --- |
| Working prototype: extract, explain, flag, search | Accuracy benchmark, behavior verification, s390x testing | Web UI, copybook and CICS/DB2 support, audit report export |

- Impact: faster, safer modernization with audit-ready traceability.

# Thank You

GitHub: https://github.com/VanshikaAsthana/cobol-rulecat

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